# THARUNKUMAR K

**DATA SCIENCE INTERN | APPLIED MACHINE LEARNING | ALGORITHMIC SYSTEMS**

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## PROFESSIONAL SUMMARY

Data Science & Applied AI practitioner with proven experience formulating statistical models, multimodal neural retrieval pipelines, and production machine learning workflows. 1st Place National Winner at Code Thugs 2K26, 3rd Prize at Neural Minds AI/ML Hackathon, and 2× SIH Top 50 Nominee. Proficient in Python, C++, SQL, NLP transformers (FinBERT, Gemini), multimodal vector embeddings (CLIP, ChromaDB), and Monte Carlo simulations. Creator of engineering platforms serving **365+ daily active users**.

## EDUCATION

**Bachelor of Technology (B.Tech) in Information Technology** — VSB Engineering College, Karur, TN **2024 – 2028**

Cumulative GPA: **8.5 / 10.0** | Expected Graduation: **May 2028**

## TECHNICAL SKILLS

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| **Data Science & ML:** | Machine Learning, Deep Learning, Statistical Modeling, Exploratory Data Analysis (EDA), Hypothesis Testing, Monte Carlo Simulations, Feature Engineering, Model Evaluation (Precision, Recall, F1, ROC-AUC) |
| **AI, NLP & Vision:** | RAG Architectures, Multimodal Embeddings (CLIP), Transformers (FinBERT, Hugging Face), Google Gemini AI API, Ollama (Local LLaMA 3), Semantic Vector Search, ChromaDB |
| **Languages & Core:** | Python (NumPy, Pandas, Scikit-Learn, PyTorch), C++, Java 17, SQL (PostgreSQL, MySQL), TypeScript |
| **Data Systems & Tools:** | PostgreSQL, Supabase, Neon Serverless, Redis, Docker, Git, GitHub Actions, REST APIs, WebSockets |

## PROFESSIONAL EXPERIENCE

**Full-Stack & Applied AI Intern** — Neura Global **06/2026 – 07/2026**

- **Semantic NLP Screening Engine:** Built automated candidate screening pipeline using **Google Gemini AI** NLP embeddings, computing semantic job-match similarity scores across **500+ student profiles** with 80% reduction in manual evaluation latency.
- **Data Pipeline & Telemetry:** Automated eligibility qualification filters (CGPA distribution, arrears, departmental quotas) using PostgreSQL and real-time Socket.IO broadcasts with zero candidate scheduling conflicts.

**Software Engineering & Analytics Intern** — Infosys Springboard **11/2025 – 01/2026**

- **Workforce Analytics Platform:** Developed **StarWall** peer-achievement and organizational analytics dashboard in Java 17 with modular microservices and relational SQL data models under strict Test-Driven Development (TDD).

## FEATURED DATA SCIENCE & AI PROJECTS

**FINDORA AI — Autonomous Multimodal Correlation Network** | *3rd Prize AIML Hackathon* **2026**

*Python*, *TypeScript*, *React 19*, *Node.js*, *Express*, *Supabase PostgreSQL*, *Google Gemini 2.5 Flash* `[Demo]` | `[GitHub]`

- **Multimodal Correlation Algorithm:** Formulated weighted similarity matching model fusing **Visual (35%)**, **Semantic Text (30%)**, and **Spatial-Temporal Proximity (35%)** embeddings to autonomously correlate lost-and-found items.
- **Anti-Fraud Heuristics:** Integrated zero-knowledge verification using normalized **Levenshtein string distance** scoring and velocity rate-limiting to eliminate fraudulent claims.

**RiskLens AI — Quantitative Risk Modeling & Financial NLP** **2026**

*Python*, *PyTorch*, *FinBERT NLP*, *Monte Carlo Simulation*, *Pandas*, *NumPy*, *SEC EDGAR API* `[GitHub]`

- **Financial Sentiment Extraction:** Built NLP pipeline extracting risk sentiment from SEC 10-K filings using fine-tuned **FinBERT** transformers to quantify corporate exposure shifts.
- **Monte Carlo Value-at-Risk (VaR):** Executed parametric and historical **Monte Carlo simulations (10,000+ paths)** to model downside risk distributions and 95%/99% VaR metrics under stressed market volatility.

**AURA — Air-Gapped Multimodal RAG Retrieval Engine** **2026**

*Java 17*, *Spring Boot*, *Python*, *ChromaDB Vector Store*, *CLIP Embeddings*, *Ollama (LLaMA 3)* `[GitHub]`

- **Vector Store & Cross-Modal Indexing:** Built air-gapped RAG platform using **ChromaDB** and **CLIP multimodal embeddings** for cross-modal document text and visual image semantic retrieval with sub-second latency.
- **Local Inference Optimization:** Streamed local LLM token generation via Ollama with loopback network isolation, ensuring complete data privacy and zero API inference costs.

**VSBEC IT Vault — Algorithmic Telemetry Platform** | *Top 50 SIH Nominee (365+ Daily Users)* **2026**

*React 19*, *TypeScript*, *Node.js*, *Express*, *PostgreSQL*, *Monaco Editor*, *Docker*, *GraphQL Telemetry* `[Demo]` | `[GitHub]`

- **Algorithmic Telemetry Analytics:** Aggregated student algorithmic metrics from **LeetCode GraphQL** and **GitHub REST** APIs, computing time-series coding velocity and streak consistency for **365+ daily students**.
- **Isolated Compiler Sandbox:** Built multi-language execution runtime (C, C++, Java 17, Python 3) with timeout traps, server-side test evaluation, and hierarchical audit trails.

## HONORS & AWARDS

- **1st Place National Winner** — *Code Thugs 2K26* (National Hackathon Champion for GOAT Collaborative IDE; ₹5,000 cash prize).
- **3rd Prize Winner** — *8-Hour AI/ML Hackathon 2026* (Awarded by Dept of AI&ML for FINDORA AI multimodal network).
- **National Finalist** — *Odoo × NMIT Bangalore Hackathon 2026* (Qualified for 24-hr Grand Finale at NMIT Bangalore; Work Suite HRMS).
- **Top 50 / 300+ Teams** — *Smart India Hackathon (SIH 2026 & SIH 2025)* (Official Central SIH Portal Nominee for IT Vault & Smart Helmet).
- **National Finalist** — *India Innovates 2026* (National Innovation Showcase for applied technology).

## VERIFIED CERTIFICATIONS & CREDENTIALS

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| **Data Science & AI:** | Tata GenAI Powered Data Analytics (Forage); Google Cloud: Responsible AI, Intro to LLMs, Vertex AI Prompt Design |
| **Software & Systems:** | Infosys Springboard Internship Certificate; Cisco Introduction to IoT (Credly); TCS iON Career Edge |